CS/ECE 4501 / CS 6501 / ECE 6782 **Due by Oct.7** 11 : 59 : 59**PM**

Problem Set II

1. (CS 6501 / ECE 6782 30%) (CS/ECE 4501 40%) **Euler's method:** `https://en.wikipedia.org/wiki/Euler_method`

   For $\frac{dy}{dt} + 2y = 2 - e^{-4t}, y(0) = 1$,

   (a) Derive its closed-form solution on your own.

   (b) Use Euler's Method to find the approximation to the solution at $t = \{1, 2, 3, 4, 5\}$ using 5 time steps, and compare to the exact solution in (a) by plotting them on a same figure.

   (c) Use different step size $h = \{0.1, 0.05, 0.01, 0.005, 0.001\}$ and plot out your approximated function value.

2. (CS 6501 / ECE 6782 50%) (CS/ECE 4501 60%) **Generating time-sequential images with geodesic shooting equations.** Given an initial velocity v_0 and a source image s (included in the "PS2-Q2" folder), implement geodesic shooting algorithm (via Euler integration) to generate a time-sequence of deformations (transformations), $\phi_t, t \in [0, \cdots, 1]$. You will then deform a given source image (included in the data folder) by using the final transformation ϕ_1 at time point $t = 1$.

   Note that the initial condition v_0 is given in the data folder, and the initial condition for ϕ_0 is an image coordinate of the given source image, which can be easily generated from Python (e.g., numpy's meshgrid).

   Your task will be:

   (a) Implement and compute the geodesic shooting equation below (a special case of the original shooting equation discussed in class) using Euler integration with 10 time steps, e.g., $t = \{0, 0.1, 0.2, \cdots, 1\}$.

   $$\frac{dv_t}{dt} = -K[(Dv_t)^T \cdot v_t + (Dv_t) \cdot v_t + v_t \cdot \text{div}(v_t)],$$
   $$\frac{d\phi_t}{dt} = v_t \circ \phi_t,$$

   where K is a smoothing kernel (e.g., a Gaussian smoothing kernel), D is a Jacobian matrix, div is a divergence operator, and $\circ$ denotes an interpolation.

   (b) Report the final velocity $v_{t=1}$ and a time-sequence of deformed source image s by computing $s \circ \phi_t$.

   (c) Report the total running time of your shooting algorithm.

   (d) Now feel free to find any online image that includes simple objects (e.g., animal shapes, human faces, or anything that you'd like to deform/warp) and use it as the source image s. Then, generate your own random initial velocity field by computing $v_0 = \epsilon \cdot \nabla s$, where ϵ is random noises sampled from a Gaussian distribution $N(\mu, \sigma^2)$ and then smoothed with a Gaussian kernel. To see the different deforming effect on the images, you may try different parameters of μ and σ. Repeat (a)-(b). Did you find anything that is interesting? Describe your findings.

* Interpolation function: Python function *scipy.ndimage.map_coordinates* with the option 'order=3'.

3. (**CS 6501 / ECE 6782 ONLY.** 20%) Compute the inverse transformation ϕ_1^{-1} at time point $t = 1$ by the following strategy.

$$\frac{dv_t}{dt} = K[(Dv_t)^T \cdot v_t + (Dv_t) \cdot v_t + v_t \cdot \mathrm{div}(v_t)],$$
$$\frac{d\phi_t^{-1}}{dt} = -D\phi_t^{-1} \cdot v_t.$$

Report the deformed source image using ϕ_1^{-1}, and describe how it is different from the final deformed image in problem 2.

IMPORTANT NOTES:

* All results should be clearly reported and discussed in the report.

* This is not required, but students in CS/ECE 4501 are welcome to use Q3 for bonus points.